Predicting Student Academic Success Using Statistical Modeling and Machine Learning

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1. Introduction

Early identification of students at academic risk is a core component of learning analytics and institutional assessment. This project builds a reproducible analytics pipeline to predict whether a student will **pass or fail** based on information available prior to course completion.

The workflow demonstrates skills aligned with assessment and analytics work: data preparation, SQL database management, statistical modeling, supervised machine learning, visualization, and reporting.

2. Data Description

This analysis uses the public UCI student performance dataset (N = 395). Key variables include demographics (e.g., sex, age, school), engagement (e.g., absences, study time, prior failures), and early academic performance (G1).

> **Outcome Variable:** `pass_fail` (1 = G3 ≥ 10, 0 = G3 < 10)
> To avoid data leakage, the final grade (G3) was used only to construct the target variable and was excluded from predictors.

The dataset was loaded into a relational SQLite database with three tables: `students`, `engagement`, and `outcomes`. SQL joins were used to create the modeling dataset and to compute assessment-style KPIs.

3. Methodology

3.1 Data Preparation

- Loaded raw CSV data and created a relational schema in SQLite
- Standardized and renamed variables (e.g., parental education features)
- One-hot encoded categorical features (school, sex, address)
- Standardized numeric predictors
- Train/test split: 75% / 25% (stratified)

3.2 Exploratory Data Analysis

Visualizations were generated using matplotlib to explore performance distribution and engagement patterns. These plots reflect common reporting outputs in learning analytics contexts.

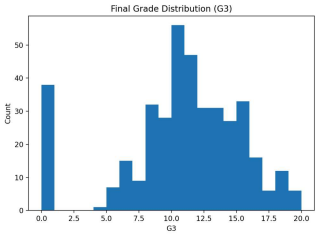

Figure 1. Distribution of final grades (G3).

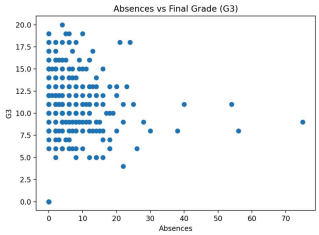

Figure 2. Relationship between absences and final grade (G3).

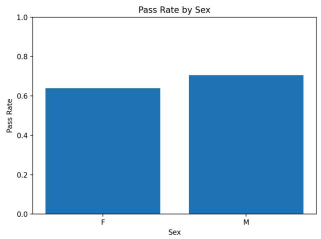

Figure 3. Pass rate by sex (example subgroup KPI).

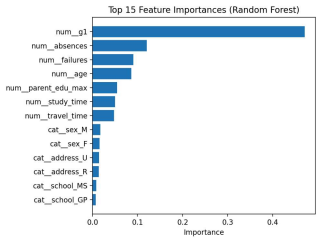

Figure 4. Random Forest feature importance (top predictors).

3.3 Models

Two supervised learning models were trained to predict `pass_fail`:

- **Logistic Regression** (interpretable baseline common in institutional research)
- **Random Forest** (nonlinear model with feature-importance output)

Performance was evaluated using **ROC-AUC** on the test set. ROC-AUC measures how well the model separates students who pass versus fail (0.50 = random, 1.00 = perfect).

4. Results

Model	ROC-AUC (Test Set)	Interpretation
Logistic Regression	**0.905**	Strong discrimination; interpretable relationships between predictors and pass/fail.
Random Forest	**0.880**	Strong performance; useful for identifying key predictive signals via feature importance.

Both models achieved strong predictive performance. Logistic regression slightly outperformed the random forest, which suggests that the relationship between predictors and student success may be largely linear and that an interpretable model may be sufficient for decision support.

Key Predictors

Feature importance analysis indicates that the strongest predictors of pass/fail status include:

- First period grade (G1)
- Prior failures
- Absences
- Study time

5. Implications for Learning Analytics

The results support the feasibility of an early-warning approach that uses early performance and engagement signals to identify students who may benefit from timely support. In an institutional context, these results could inform:

- Early outreach when absences exceed a threshold
- Targeted tutoring or advising for students with low early grades
- Monitoring dashboards for program assessment and retention initiatives

6. Reproducibility & Workflow

The project is structured as a reproducible workflow with clear separation of data, SQL schema/queries, processing scripts, model outputs, and report artifacts.

- `src/build_db.py` creates the SQLite database from raw CSV
- `src/eda.py` generates descriptive statistics and EDA plots

- `src/train_model.py` trains models, evaluates ROC-AUC, and saves metrics/feature importance plots
- Outputs saved to `outputs/` (figures and metrics JSON)

7. Conclusion

This project demonstrates applied learning analytics using SQL and Python for data preparation, modeling, evaluation, and reporting. The strong ROC-AUC results indicate that early academic performance and engagement variables can provide meaningful predictive signal for student success.

Project artifacts: SQLite database, SQL queries, reproducible Python scripts, figures, and metrics.